

Machine Learning Models for Heart Disease Prediction: Performance, Robustness, and Fairness

Performance Comparison on the Cleveland Heart Dataset

The Cleveland Heart Disease dataset (a UCI repository subset) has been a benchmark for comparing ML models in cardiovascular risk prediction. Traditional **logistic regression (LR)** was one of the first methods applied; for instance, Detrano et al. (1989) reported about **77% accuracy** with a basic logistic model ¹ ². Recent studies show that with improved preprocessing and feature scaling, even simple LR can reach much higher performance – one 2025 experiment achieved **~92.3% accuracy** using LR ³. Nonetheless, modern non-linear models typically outperform linear models on this task. In an extensive 2025 evaluation of 21 algorithms, **XGBoost** (an ensemble tree boosting method) yielded the best results, with roughly **97-98% accuracy and AUC ~0.98**, surpassing many other classifiers ⁴. **Random Forests (RF)** and other tree ensembles also consistently rank among top performers ⁵ ⁴. For example, using the Cleveland data, an untuned RF achieved about **84% accuracy (AUC ~92%)**, while an untuned XGBoost got ~82% (AUC ~91%); after advanced hyperparameter tuning, RF and XGBoost rose to **93-95% accuracy** on the same data ⁵. **Support Vector Machines (SVM)** show moderate-to-high performance depending on the kernel: using a non-linear radial basis function (RBF) kernel on Cleveland, SVM reached about **78.5% accuracy** versus ~70% with a linear kernel ². Classic **k-Nearest Neighbors (KNN)** and **Naïve Bayes** tend to perform a bit lower (often in the 70-85% range in literature) unless combined with feature engineering ⁶. Meanwhile, results with **neural networks** have been mixed on this tabular dataset. Simpler feed-forward networks (ANNs) can achieve ~80-90% accuracy ⁶, and one study using PCA for dimensionality reduction reported **95.5% accuracy** with a neural network ⁷. However, deep learning doesn't always dominate here – in one comparison a deep neural net obtained ~81.9% accuracy, no better than SVM's 86.2% on the same data ⁶. Overall, tree-based ensemble models (Random Forest, XGBoost, Gradient Boosting) have shown **highest predictive power** for heart disease risk, often outperforming single classifiers. Complex hybrids (e.g. stacking or optimized ensembles) can further improve accuracy, but even well-tuned “simple” models like LR or SVM can approach 85-90%+, especially with careful data preprocessing ³ ⁸.

Robustness to Imbalanced or Overlapping Classes

In medical datasets, class imbalance and overlapping feature distributions are common challenges. The Cleveland dataset is roughly balanced (165 patients with disease vs 138 without ⁹), but many real-world cohorts have fewer positive cases. Generally, **no classifier is immune to imbalanced data** – without adjustment, most will favor the majority class. Techniques like class weighting or resampling are often needed regardless of model. That said, some models inherently cope better once such techniques are applied. **Ensemble methods** (especially boosting like XGBoost) tend to be robust in the sense that they focus on misclassified examples – often the minority class – which can improve minority class detection. For example, XGBoost's iterative boosting gives it an edge in handling class imbalance, as it increasingly prioritizes hard-to-classify cases. **Decision tree-based models** can handle overlapping feature space by learning non-linear splits; a Random Forest can partition features in many ways to capture complex decision boundaries that separate classes even when their raw distributions overlap.

This often lets tree ensembles outperform linear models in scenarios with overlapping class clusters. By contrast, a **logistic regression**, with its single linear decision boundary, may struggle if the classes are not linearly separable. In such cases, non-linear kernels or feature transformations are required to improve LR or SVM performance ². SVM with an RBF kernel, for instance, showed much higher accuracy (78% vs 70%) than a linear SVM on the Cleveland task by better handling non-linear overlaps ². **Neural networks** are universal function approximators and in theory can model very overlapping classes, but in practice they need enough data and tuning to do so. On small tabular datasets like Cleveland (only a few hundred samples), a complex NN can overfit and may not generalize better than simpler models ⁶. In terms of **robustness**, tree ensembles and kernel SVMs offer flexibility to handle overlapping distributions, whereas linear models are less forgiving without extensive feature engineering. For **class imbalance**, boosting and bagging methods can integrate sampling strategies (e.g. class-weighted trees or balanced mini-batches) more easily, but ultimately any model can be made to address imbalance via data-centric approaches (oversampling minority cases, generating synthetic data, etc. ¹⁰). A 2022 study underscores the importance of data quality in this context: it found that removing outlier “anomalous” records from the heart disease dataset significantly **improved the accuracy of multiple models (RF, SVM, LR)** on that data ¹¹. This implies that cleaning noisy, overlapping points benefits all models’ robustness. In summary, ensemble models (Random Forest, XGBoost) combined with proper handling of class imbalance are considered **most resilient** to skewed or overlapping class distributions, while simpler models require more careful data preprocessing to reach comparable performance.

Explainability and Fairness Considerations

Explainability is a crucial factor in deploying ML for healthcare. Simpler algorithms like logistic regression and decision trees are *inherently interpretable*: LR provides weights for each risk factor, and a decision tree offers a clear rule structure. These transparent “glass-box” models enable clinicians to understand **why** a prediction was made, which in turn can foster greater trust and accountability ¹². In fact, a recent review emphasizes developing interpretable models (or using explanation tools) to improve provider trust – highlighting that black-box models can lead to confusion or hidden biases in medical decisions ¹³. By contrast, complex ensemble and deep learning models are **opaque**. Random forests, XGBoost, SVM with non-linear kernels, and neural networks all act as black boxes where the decision logic is not human-readable. To gain insight into these, practitioners must apply *post-hoc* explainability techniques. Methods like **LIME** and **SHAP** can provide model-agnostic explanations – for example, indicating which clinical features most influenced a particular prediction ¹⁴. Such tools can indeed uncover if a model is relying on inappropriate signals or exhibiting bias ¹⁵. Nevertheless, there is a growing view in healthcare AI that using inherently interpretable models where possible is preferable to solely relying on explainers for black-boxes ¹³. This is especially relevant for **fairness** and ethical AI. An explainable model makes it easier to detect bias – e.g. if a risk model were unfairly giving higher risk scores to a certain demographic, a transparent model would allow stakeholders to see that pattern and correct it. A 2025 study noted that **Explainable AI (XAI) techniques play a major role in ensuring fairness** in clinical ML by addressing transparency and helping avoid bias ¹⁶.

Fairness in cardiovascular risk models also depends heavily on the *data*. Bias often originates from unbalanced or unrepresentative data – for instance, if the training data had mostly male patients, a model might perform worse for females. Simply changing the model type cannot solve such bias; the literature stresses the need for **diverse, representative datasets** and even augmentation of minority groups to mitigate bias ¹⁰. This is where **Data-Centric AI** insights come in: rather than exclusively tweaking models, one should improve data quality and coverage. Ensuring all relevant populations are well-represented, addressing missing values, and correcting mislabeled or noisy entries can greatly improve both performance and fairness ¹⁷. Andrew Ng and others advocate this data-centric approach – treating high-quality data as the cornerstone, since models are only as fair and accurate as the data

they learn from ¹⁸. For example, one study combined four heart disease subsets and balanced the data, yielding robust results across subgroups when using advanced models ¹⁹ ²⁰. Techniques like removing biased features, or using fairness-constrained algorithms, can be applied to any model, but they are often easier to implement and audit in simpler models. Notably, “fairness through unawareness” (simply excluding sensitive attributes like sex or race) is not a guarantee of fairness – models can still pick up proxy variables ²¹. Instead, fairness-aware modeling might involve enforcing equity constraints or post-processing outputs to equalize opportunity between groups ²², which can be done with any classifier but again is more straightforward when the model is interpretable.

In summary, **model choice affects explainability and the ease of bias mitigation**. Logistic regression and small decision trees offer **better explainability** out-of-the-box and thus more direct insight into fairness issues. Ensemble methods and deep networks often **perform better** on predictive accuracy but require extra diligence – using XAI tools and careful validation – to ensure their decisions are fair and transparent ¹³ ¹⁶. A balanced approach informed by data-centric principles is recommended: improving data quality and balance (thus reducing inherent bias) and selecting models appropriate for the clinical context (sometimes a slightly less complex but interpretable model may be favored for ethical reasons ¹³). By doing so, one can achieve high accuracy in predicting cardiovascular risk **while** maintaining robustness, explainability, and fairness in the model’s predictions.

Sources: Recent comparative studies and reviews in cardiovascular ML ⁵ ⁴ ² ¹³, healthcare AI fairness literature ¹⁰ ¹⁶, and data-centric AI insights ¹¹ ¹⁷.

¹ [PDF] Comparative Study of Classification Techniques (SVM, Logistic ...
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² ⁶ ⁷ ⁹ ¹¹ ¹⁹ The Efficacy of Machine-Learning-Supported Smart System for Heart Disease Prediction
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³ ⁸ Effectiveness of machine learning models in diagnosis of heart disease: a comparative study | Scientific Reports
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⁴ ¹⁶ ²⁰ An extensive experimental analysis for heart disease prediction using artificial intelligence techniques | Scientific Reports
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¹⁰ ¹² ¹³ ¹⁴ ¹⁵ AI-driven healthcare: A review on ensuring fairness and mitigating bias | PLOS Digital Health
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¹⁷ ¹⁸ Data-Centric AI // van der Schaar Lab
<https://www.vanderschaar-lab.com/data-centric-ai/>

²¹ ²² M04_Bias_and_Fairness.pdf
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